Early Detection of Diabetes Risk Using Predictive Modeling

OMIS 3020

March 25, 2026

By: Aditya Batra, Liza Berin, Vasilios Flegas, Yu Qing Koh

Problem Overview

Diabetes as a Growing Health Challenge

Diabetes rates are increasing globally

Driven by aging population and lifestyle factors

Late diagnosis leads to serious complications

Heart disease, kidney failure, reduced quality of life

Healthcare systems face increasing pressure

Rising costs and demand for treatment

Opportunity: Early risk prediction using data

Identify high-risk individuals before diagnosis

Research Questions

1. Can predictive models accurately classify individuals with diabetes or no diabetes based on the risk factors?

2. Which factors are strongly associated with diabetes risk?

3. How do different machine learning models compare in terms of performance when predicting diabetes risk?

Dataset Overview

Dataset Summary

4,303 observations
18 variables
Binary target: Diabetes (Yes/No)

Variable Types

Quantitative: BMI, glucose, cholesterol, blood pressure
Categorical: gender, smoking, alcohol, family history

Key Features

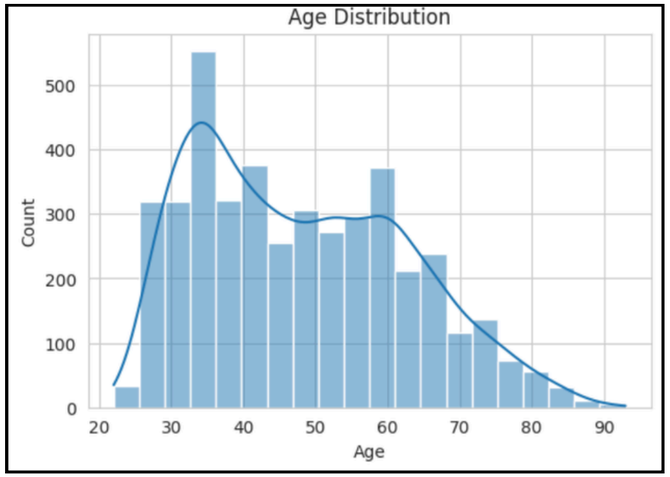
Glucose level
BMI
Blood pressure
Lifestyle behaviors

Source

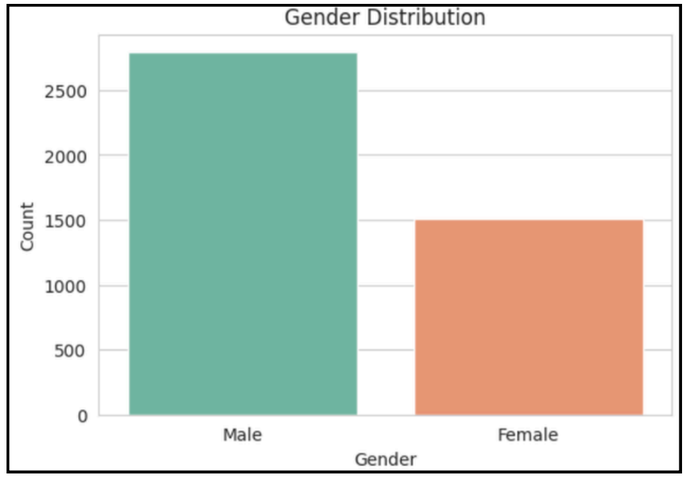
Kaggle (pkdarabi, 2023)
Public dataset (no personal identifiers)

	Age	Gender	BMI	SBP	DBP	FPG	Chol	Tri	HDL	LDL	ALT	BUN	CCR	FFPG	smoking	drinking	family_history	Diabetes
0	26	1	20.1	119	81	5.80	4.36	0.86	0.90	2.43	12.0	5.40	63.8	5.40	Never	Never	0	0
1	40	1	17.7	97	54	4.60	3.70	1.02	1.50	2.04	9.2	3.70	70.3	4.10	Current	Current	0	0
2	40	2	19.7	85	53	5.30	5.87	1.29	1.75	3.37	10.1	4.10	61.1	4.85	Never	Never	0	0
3	43	1	23.1	111	71	4.50	4.05	0.74	1.27	2.60	36.5	4.38	73.4	5.30	Ever	Never	0	0
4	36	1	26.5	130	82	5.54	6.69	3.49	0.91	3.64	69.3	3.86	67.5	5.53	Never	Never	0	0

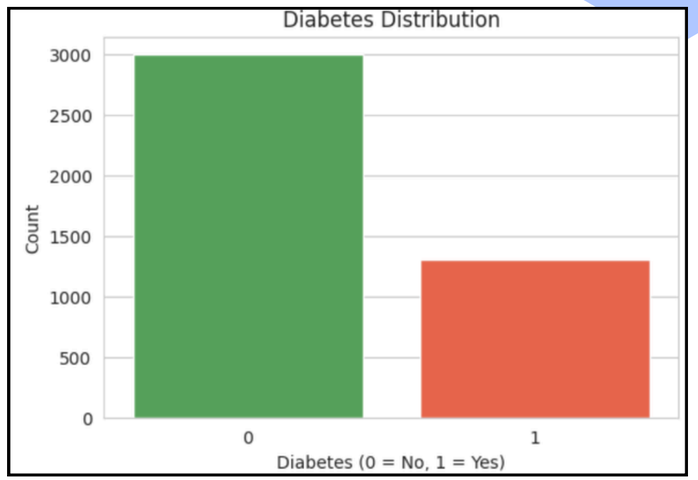
Data Demographics



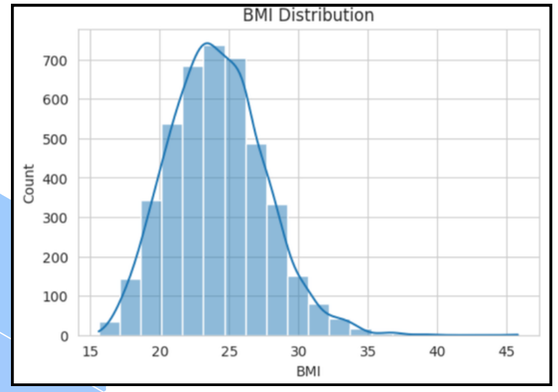
Most individuals are middle-aged, with fewer younger and older participants



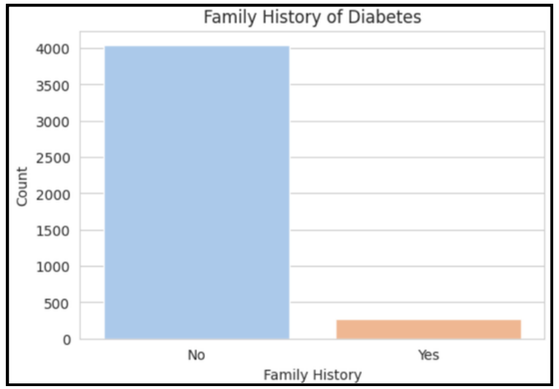
Dataset contains more males than females (65%, 35%)



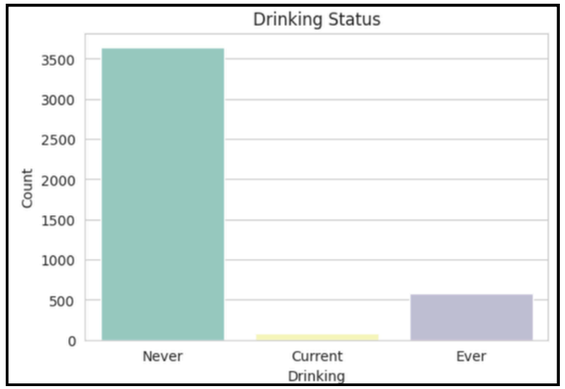
Dataset is imbalanced, with more non-diabetic cases



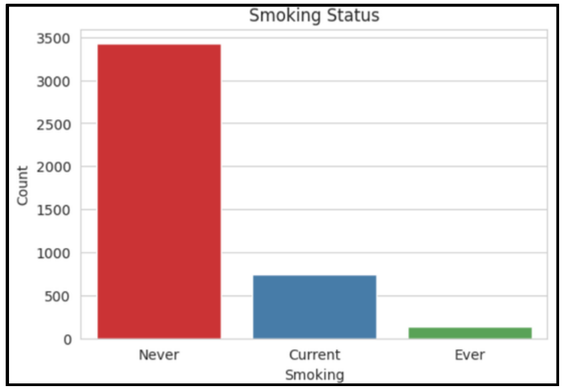
BMI is concentrated around the mid-range, with some higher outliers



Most individuals do not report a family history of diabetes



Majority of individuals report never drinking



Most individuals are non-smokers

Data Preparation

Data Cleaning

Checked missing values (df.isna())

Replaced invalid values with NaN

Standardized smoking & drinking categories

Exploratory Analysis

Used df.describe() + correlation heatmap

Insights shown in next slide

Train-Test Split

Split into X (features) and y (target)

70/30 split using train_test_split()

Stratified to preserve class balance

Feature Preparation

Separated categorical & numerical variables

Applied OneHotEncoder

Combined encoded + numerical features

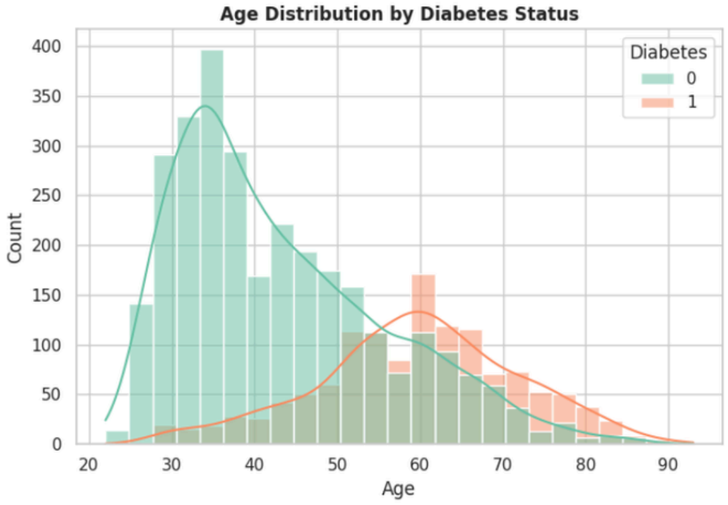
Feature Scaling

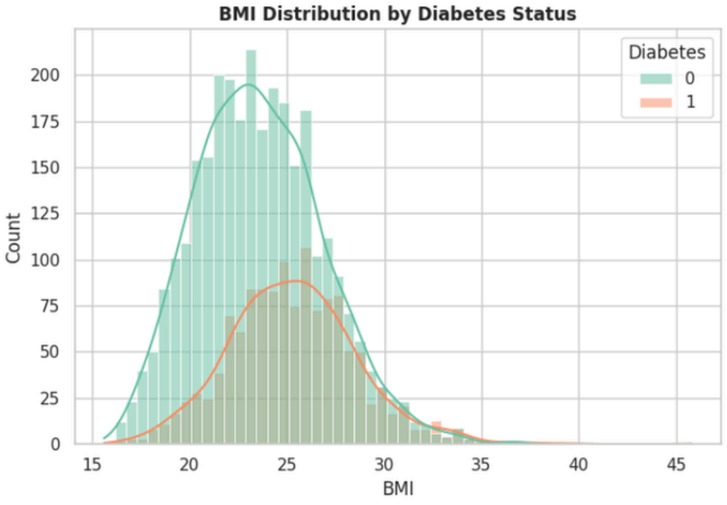
Adjusted features to the same scale

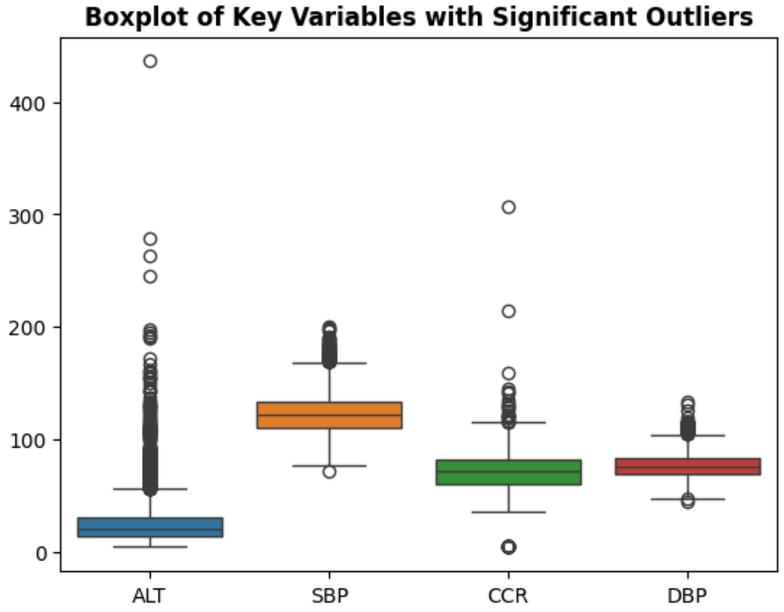
Prevented larger values from dominating

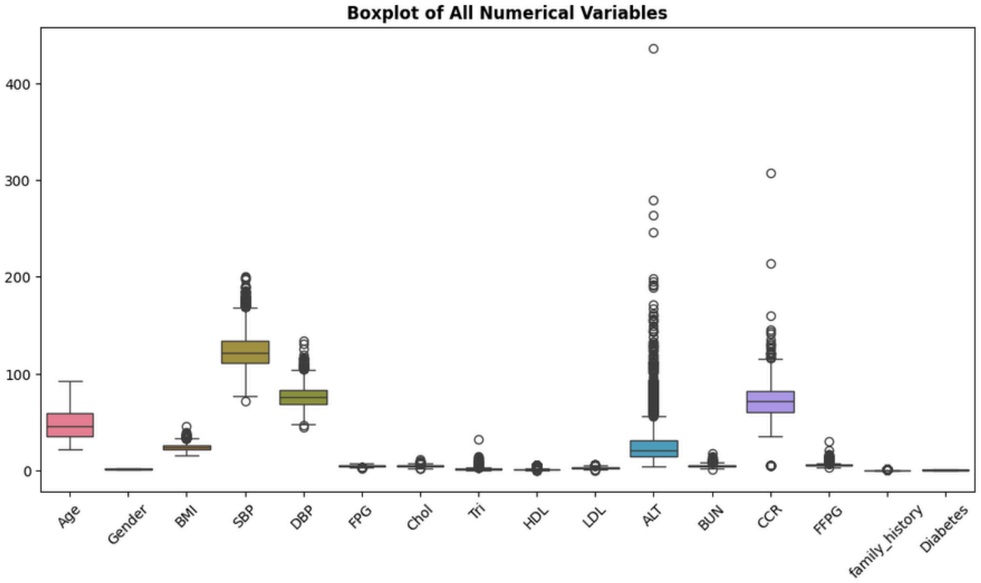
Improved model performance

Data Visualizations & Insights

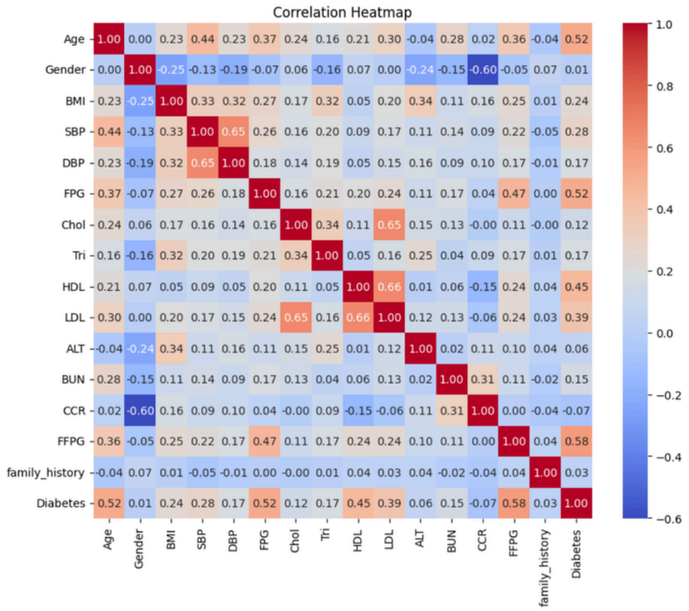


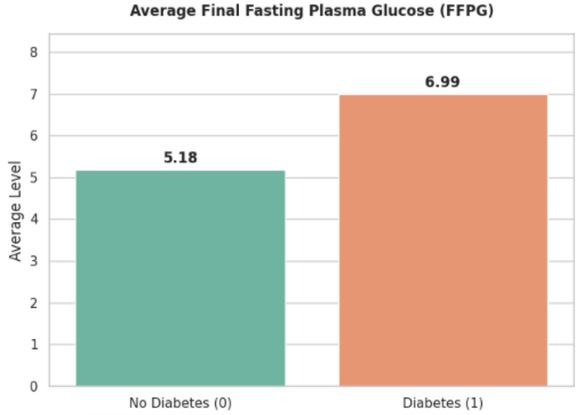


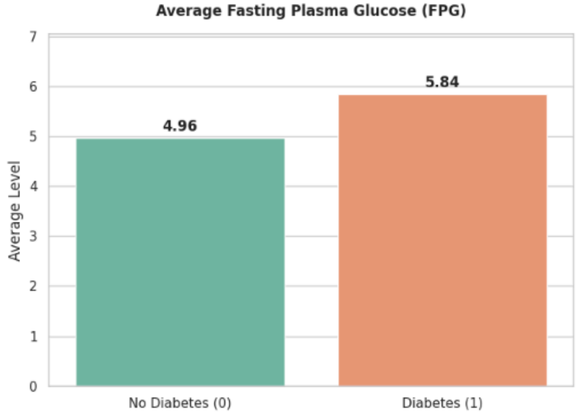


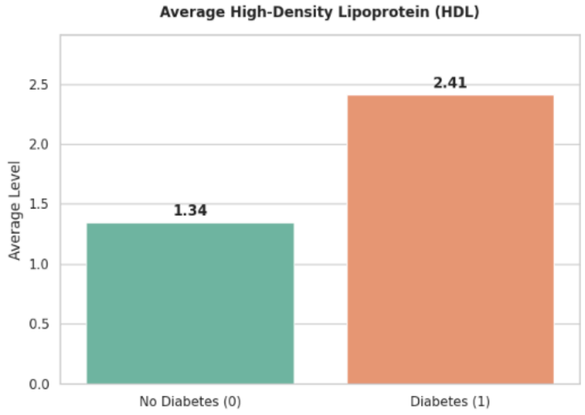


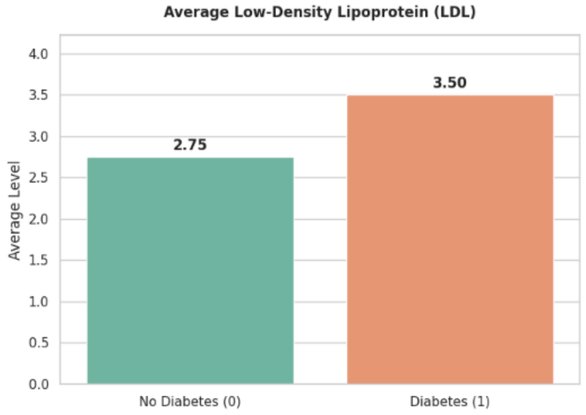
Data Visualizations & Insights











Models Used

We applied multiple classification models to compare performance and identify the most effective approach for predicting diabetes risk.

Classification

Logistic Regression

Baseline classification model

Predicts probability of diabetes

Interpretable and efficient

Decision Tree

Rule-based model (if/then splits)

Easy to interpret

Captures non-linear relationships

K-Nearest Neighbors

Classifies based on nearest data points

Uses distance (similarity)

Sensitive to feature scaling

Random Forest Classifier

Ensemble of multiple decision trees

Improves accuracy and reduces overfitting

Captures complex patterns

Support Vector Classifier

Finds optimal boundary between classes

Effective in high-dimensional data

Can model non-linear relationships

Results & Model Comparison

Logistic Regression

	precision	recall	f1-score	support
No Diabetes (0)	0.91	0.96	0.94	900
Diabetes (1)	0.89	0.80	0.84	391
accuracy			0.91	1291
macro avg	0.90	0.88	0.89	1291
weighted avg	0.91	0.91	0.91	1291

Elapsed train time for Logistic Regression (sec): 0.09776139259338379

Elapsed predict time for Logistic Regression (sec): 0.0006225109100341797

Decision Tree

	precision	recall	f1-score	support
No Diabetes (0)	0.91	0.90	0.91	900
Diabetes (1)	0.79	0.80	0.79	391
accuracy			0.87	1291
macro avg	0.85	0.85	0.85	1291
weighted avg	0.87	0.87	0.87	1291

Elapsed train time for Decision Tree (sec): 0.1466083526611328

Elapsed predict time for Decision Tree (sec): 0.001058816909790039

K-Nearest Neighbors

	precision	recall	f1-score	support
No Diabetes (0)	0.89	0.97	0.93	900
Diabetes (1)	0.91	0.72	0.80	391
accuracy			0.89	1291
macro avg	0.90	0.84	0.86	1291
weighted avg	0.89	0.89	0.89	1291

Elapsed train time for KNN (sec): 0.006253480911254883

Elapsed predict time for KNN (sec): 0.1221628189086914

Support Vector Classifier

	precision	recall	f1-score	support
No Diabetes (0)	0.91	0.96	0.94	900
Diabetes (1)	0.90	0.79	0.84	391
accuracy			0.91	1291
macro avg	0.91	0.88	0.89	1291
weighted avg	0.91	0.91	0.91	1291

Elapsed train time for SVC (sec): 0.36116528511047363

Elapsed predict time for SVC (sec): 0.1657578945159912

Results & Model Comparison

Top 10 Features

	importance
FFPG	0.275141
FPG	0.139650
Age	0.117813
HDL	0.110193
LDL	0.087486
SBP	0.037147
BMI	0.031443
Tri	0.027831
ALT	0.026278
BUN	0.026207

Glucose-related variables (FFPG, FPG) are the strongest predictors of diabetes.

Age and cholesterol levels also contribute to risk.

Model Results

	Model	Accuracy	Precision	Recall	F1	Train Time	Predict Time
0	Logistic Regression	0.907823	0.888571	0.795396	0.839406	0.097761	0.000623
1	Decision Tree	0.873741	0.785000	0.803069	0.793932	0.146608	0.001059
2	K-Nearest Neighbors (KNN)	0.891557	0.906149	0.716113	0.800000	0.006253	0.122163
3	SVC	0.910147	0.900875	0.790281	0.841962	0.361165	0.165758

SVC achieved the highest overall performance, while Logistic Regression provided strong balanced results

Recall was prioritized to minimize missed diabetes cases.

Final Model + Tuning

Model Selection

Chosen Model:

Logistic Regression

Strong overall performance

Balanced precision and recall

Interpretable and efficient

```
============================================================
Logistic Regression Fine Tuned
Accuracy: 0.9078
Confusion Matrix (rows=actual, cols=pred):
[[862  38]
 [ 81 310]]

                 precision    recall  f1-score   support

No Diabetes (0)       0.91      0.96      0.94       900
   Diabetes (1)       0.89      0.79      0.84       391

       accuracy                           0.91      1291
      macro avg       0.90      0.88      0.89      1291
   weighted avg       0.91      0.91      0.91      1291

============================================================
```

Tuning Process

Model Improvement:

Tested different settings to improve performance

Selected the best-performing version based on validation results

Final Outcome

Improved model reliability
maintained strong accuracy and recall

	Model	Accuracy	Precision	Recall	F1	Train Time	Predict Time
0	Logistic Regression	0.907823	0.888571	0.795396	0.839406	0.018077	0.000601
1	Tuned Logistic Regression	0.907823	0.890805	0.792839	0.838972	0.008794	0.000546

Challenges & Trade-offs

Class Imbalance

More non-diabetic than diabetic cases

Accuracy alone can be misleading

Precision vs Recall

Higher recall → more false positives

Higher precision → more missed cases

Dataset Limitations

Data sourced from China

May limit generalizability to Canada

Model Trade-offs

Complex models → higher performance

Simpler models → more interpretable

Conclusion & Impact

Key Findings

Glucose-related variables (FFPG, FPG) are the strongest predictors of diabetes
Age and cholesterol also significantly contribute to risk

Model Outcome

Logistic Regression achieved strong and consistent performance across all metrics
Selected for its balance, interpretability, and efficiency

Real World Impact

Enables early identification of high-risk individuals
Supports data-driven, preventive healthcare decisions

Future Improvements

Use more diverse and representative datasets
Explore advanced models and hyperparameter tuning
Incorporate additional clinical and lifestyle variables

Thank you for Listening

Questions?